Project: Smart Inventory & Replenishment System

Objective

The objective of this project is to develop an intelligent inventory management and replenishment system that enables retail operations to maintain optimal stock levels across multiple stores while minimizing both stockouts and overstock situations.

The system should leverage historical sales data and predictive analytics to forecast demand and automate replenishment decisions. By doing so, it should improve product availability, reduce waste, and optimize supply chain efficiency.

Additionally, the solution should empower store managers and supply chain teams with actionable insights and intuitive tools to monitor inventory performance and intervene when necessary.

Description

Retail businesses often struggle with balancing inventory levels due to fluctuating demand, seasonal trends, and limited visibility across stores. Overstocking leads to increased storage costs and waste, while stockouts result in lost sales and poor customer experience.

This system aims to address these challenges by providing a centralized platform that tracks inventory levels in real time across all locations. It will analyze historical sales patterns, identify trends, and generate demand forecasts for each product.

Based on these forecasts, the system will automatically generate replenishment recommendations, suggesting when and how much stock should be reordered. It should also provide alerts for critical situations, such as low stock levels or excess inventory.

Users will interact with the system through a dashboard that presents key metrics, trends, and recommendations. Store managers should be able to adjust or override system suggestions based on local knowledge or unexpected events.

The system must also account for dependencies such as supplier lead times, delivery schedules, and warehouse constraints. The goal is to create a more proactive and data-driven approach to inventory management that reduces inefficiencies and improves overall business performance.

Requirements

- Track inventory levels across multiple stores

- Predict product demand based on historical data

- Generate restocking recommendations

- Provide alerts for low stock or overstock situations

- Dashboard for store managers and supply chain teams

- Allow manual adjustments to recommendations

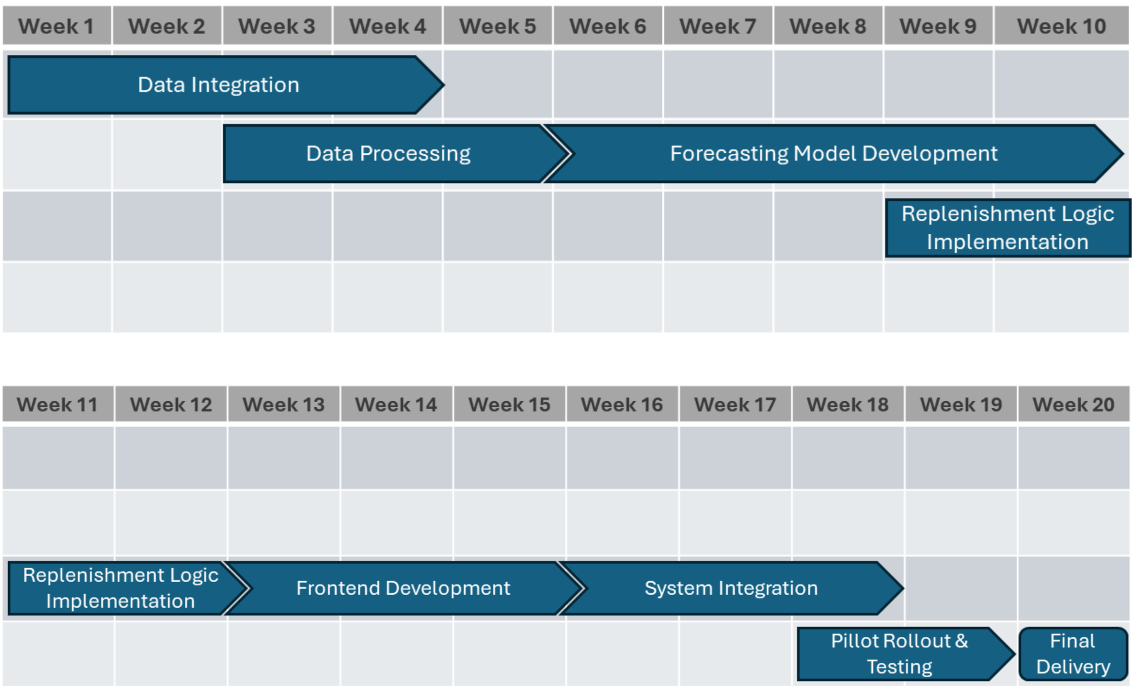
Deadlines

- MVP Delivery: Week 15

- Pilot Store Rollout: Week 20

Milestones

1. Data integration from inventory systems (Week 1-4)
2. Data preprocessing and validation (Week 3-5)
3. Demand forecasting model development (Week 6-10)
4. Replenishment logic implementation (Week 9-12)
5. Frontend dashboard development (Week 13-15)
6. System integration (Week 16-18)
7. Pilot rollout and testing (Week 18-19)
8. Final delivery (Week 20)



Dependencies

- Forecasting depends on historical sales data
- Recommendations depend on forecasting output
- Dashboard depends on backend services

Team Profiles

Core Roles

- Product Owner - defines vision, priorities, and business value
- Business Analyst - translates requirements and maps inventory processes
- Supply Chain Specialist - defines replenishment logic and constraints (lead times, stock policies)

Technical Roles

- Backend Developer - builds APIs and inventory/replenishment logic
- Frontend Developer - develops dashboard and user interface
- Data Scientist / ML Engineer - builds demand forecasting models
- Data Engineer - handles data integration, preprocessing, and pipelines

Support Roles

- UX/UI Designer - designs dashboards and user experience
- DevOps Engineer - manages infrastructure and deployments